

Basic Principles of Medicine 1
Module: Foundation to Medicine | Lecture 1

Histology of the Cytoskeleton of the Cell

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Objectives

List the three domains of cellular life.	SOM.MK.I.BPM1.1.FTM.3.HCB.0709
List the three major types of protein filaments (cytoskeletal elements) that form the cytoskeleton.	SOM.MK.I.BPM1.1.FTM.3.HCB.0710
State the function(s) of microtubules as a cytoskeletal element.	SOM.MK.I.BPM1.1.FTM.3.HCB.0711
State the function(s) of intermediate filaments as a cytoskeletal element.	SOM.MK.I.BPM1.1.FTM.3.HCB.0712
State the function(s) of microfilaments as a cytoskeletal element.	SOM.MK.I.BPM1.1.FTM.3.HCB.0713
Describe the protein structure of microtubules as a cytoskeletal element.	SOM.MK.I.BPM1.1.FTM.3.HCB.0714
Describe the protein structure of intermediate filaments as a cytoskeletal element.	SOM.MK.I.BPM1.1.FTM.3.HCB.0715
Describe the protein structure of microfilaments as a cytoskeletal element.	SOM.MK.I.BPM1.1.FTM.3.HCB.0716
Describe the assembly of microtubules.	SOM.MK.I.BPM1.1.FTM.3.HCB.0717
Describe the assembly of intermediate filaments.	SOM.MK.I.BPM1.1.FTM.3.HCB.0718
Describe the assembly of microfilaments.	SOM.MK.I.BPM1.1.FTM.3.HCB.0719
Discuss three compounds that can affect microtubule function.	SOM.MK.I.BPM1.1.FTM.3.HCB.0720
Discuss two compounds that can affect microfilament function.	SOM.MK.I.BPM1.1.FTM.3.HCB.0721
State the function(s) of the centrosome.	SOM.MK.I.BPM1.1.FTM.3.HCB.0722
Describe the structure of the centrosome.	SOM.MK.I.BPM1.1.FTM.3.HCB.0723
Give the location of the basal body.	SOM.MK.I.BPM1.1.FTM.3.HCB.0724
State the function(s) of the basal body.	SOM.MK.I.BPM1.1.FTM.3.HCB.0725
Describe the structure of the basal body.	SOM.MK.I.BPM1.1.FTM.3.HCB.0726
Differentiate between the following apical specializations based on function, tissues found and their microanatomical structures: primary cilia, motile cilia and flagella.	SOM.MK.I.BPM1.1.FTM.3.HCB.0727
Differentiate between the following apical specializations based on function, tissues found and their microanatomical structures: microvilli, stereocilia.	SOM.MK.I.BPM1.1.FTM.3.HCB.0728

Objectives

Describe the terminal web.	SOM.MK.I.BPM1.1.FTM.3.HCB.0729
Describe the motor protein associated with microfilaments.	SOM.MK.I.BPM1.1.FTM.3.HCB.0730
Describe the two motor proteins associated with microtubules.	SOM.MK.I.BPM1.1.FTM.3.HCB.0731
Describe role of the cytoskeleton in intracellular transport.	SOM.MK.I.BPM1.1.FTM.3.HCB.0732
Define actin polymerization.	SOM.MK.I.BPM1.1.FTM.3.HCB.0733
Describe the three types of membrane protrusion structures.	SOM.MK.I.BPM1.1.FTM.3.HCB.0734
Describe the role of actin polymerization in membrane protrusion.	SOM.MK.I.BPM1.1.FTM.3.HCB.0735
Describe the role of actin polymerization in cellular motility.	SOM.MK.I.BPM1.1.FTM.3.HCB.0736
Explain how a cell moves across a substrate by utilizing the concept of actin polymerization.	SOM.MK.I.BPM1.1.FTM.3.HCB.0737
Describe the five types of intracellular inclusions	SOM.MK.I.BPM1.1.FTM.3.HCB.0758

Recommended Reading

- *Material covered in lecture handout

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 2: Cell Cytoplasm Pg. 62-81, Chapter 3: The Cell Nucleus Pg. 82-105

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=257424028&bookid=3290>

Lippincott's Illustrated Reviews: Cell & Molecular Biology 3e

Chandler & Viselli 2024

Chapter 4

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=255703694&bookid=3244>

Important note

Some of the slides provided in this handout will have a green background

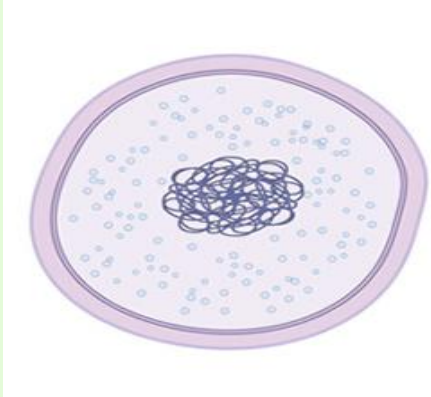
This is information that will help you understand the lecture material but will NOT be covered during the lecture delivery

Terminology

- **Cytoskeleton** – a network of proteins that provide scaffolding or allow movement for the structures inside the cell
- **Polymerization** – the process by which cytoskeletal filaments are formed and lengthened through assembly of their protein subunits
- **Treadmilling** – continual process of assembly on one end and disassembly on the other to maintain the optimal length of the cytoskeletal filament
- **Mitosis** – cell division resulting in two cells identical to the original
- **Axoneme** – the central portion (core) of cilia, flagella and primary cilia
- **Apical surface of cell** – the portion of the cell that is not attached to anything also referred to as the free surface
- **Basal surface of cell** – the portion of the cell that is attached to the basement membrane
- **Lateral surfaces of cell** – the surfaces of the cell that are adjacent to other cells
- **Terminal web** – a network of actin filaments that sit just deep to the plasma membrane
- **Signaling molecule** – a substance that can be recognized by cell surface receptors and serve as a way for cells to communicate with each other
- **Receptor** – a protein that is capable of binding to a signaling molecule. May be part of the cell membrane or in the cytosol

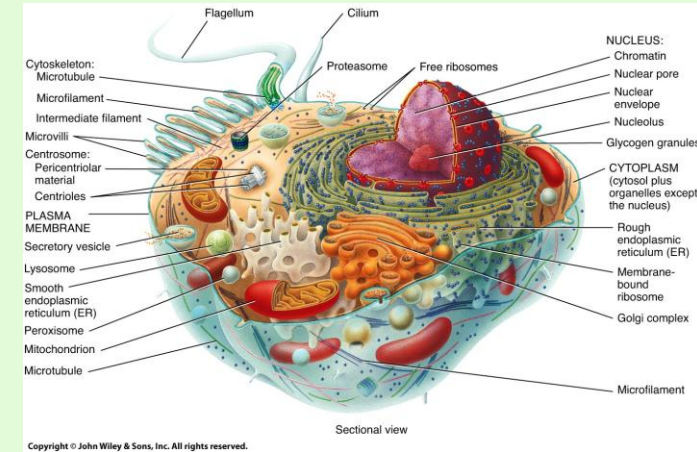
Cellular domains of life

Prokaryotes



- Single celled organisms
- Small amount of DNA
- Plasmids for adaptation
- No nucleus or organelles
- E.g.
 - Bacteria
 - Archaea - extremophiles

Eukaryotes



- Multicellular organisms
- Cells contain nucleus – for DNA
- Contain organelles with specific functions
- Cells are adapted to specific function as part of the whole

Eukaryotic Cell – Overview

- **Plasma membrane** - Outermost organelle which keeps everything together, also surrounds most organelles
- **Cytoplasm** – Everything on the intracellular aspect of the plasma membrane
 - Cytoplasmic matrix (**cytosol**) – aqueous gel containing molecules of different size where the physiological processes of the cell takes place
- **Cytoskeleton** – polymerized proteins that serve as a framework for cell structure and organization, allows for movement within the cell and the cell itself. Determines the fluidity of the cytosol and plays an important part in cell division
- **Organelles** – little organs
- **Inclusions** – accumulations in cells byproducts of metabolism

Why study cells?

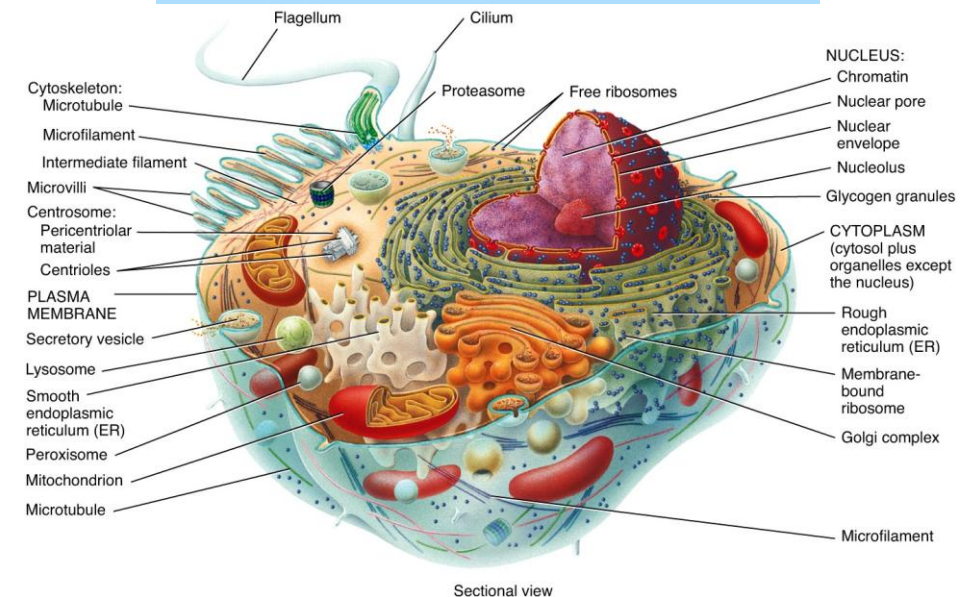
- All functions associated with life happens at a **cellular level**



- Each **organelle** has a function within the **cell**
 - Organelle numbers and location determines cell function

- Ingestion and absorption of metabolites
- Digestion and excretion of wastes
- Reproduction
- Protection
- Movement
- Death

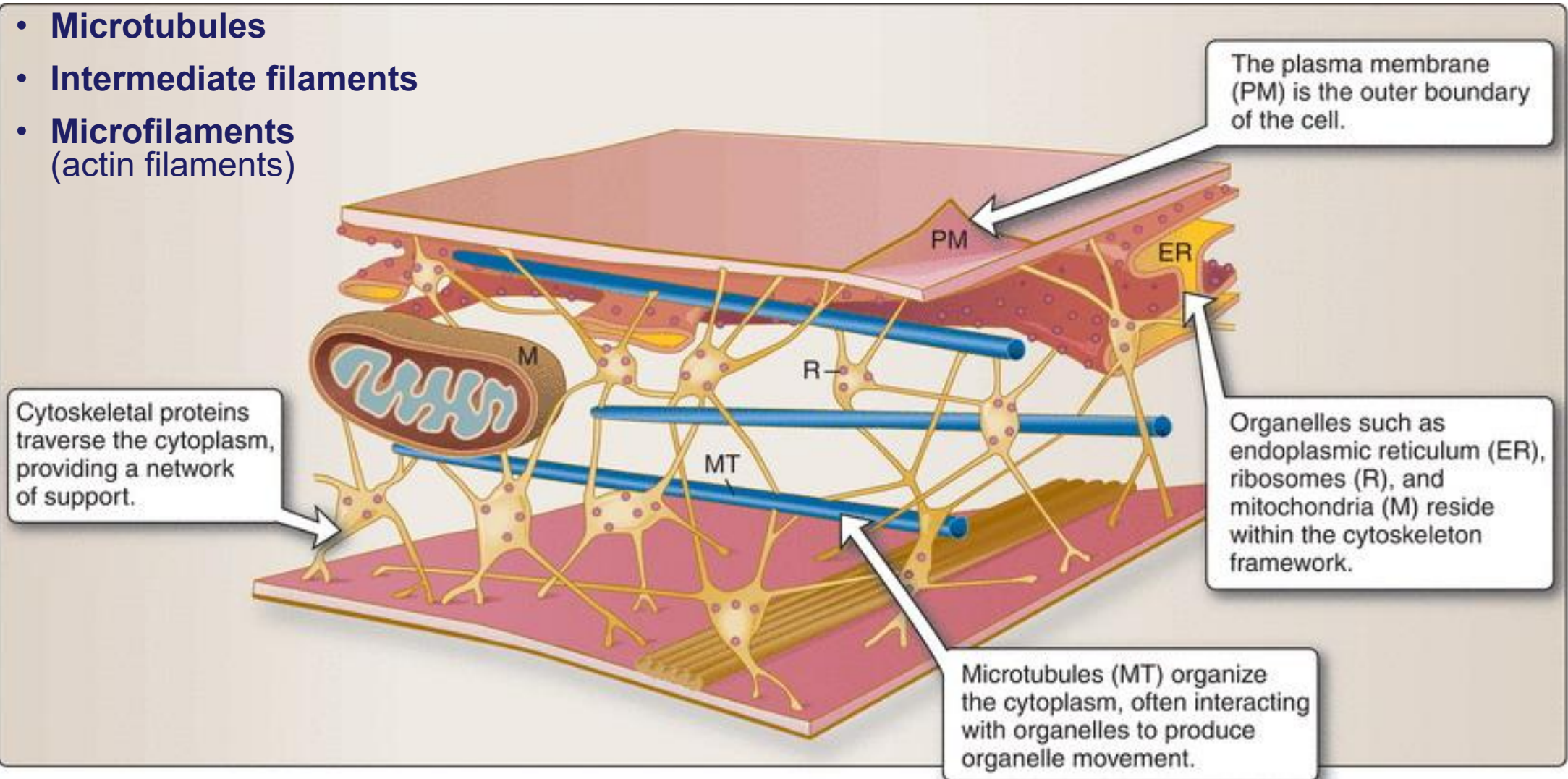
- Each **cell** has a particular function within a **tissue**
- Each **tissue** type has a function in the **organ**
- Each **organ** has a function as part of a **system**
- **Nothing in the cell happens by chance**



Cytoskeleton - Overview

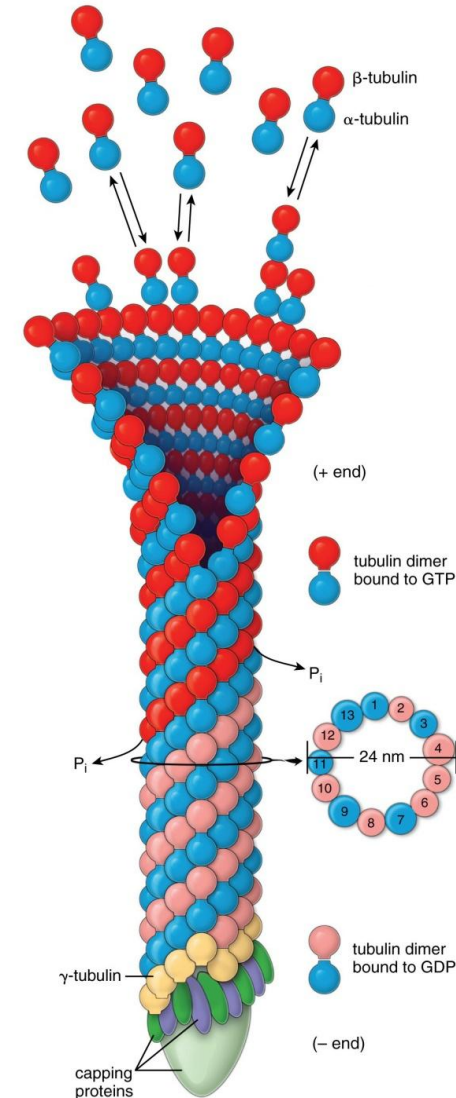
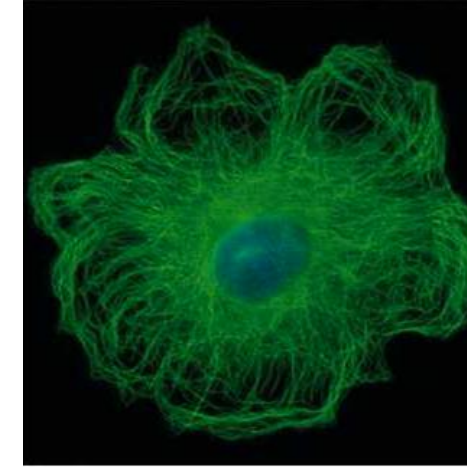
Framework for cell structure and organization that allows for movement within the cell and the cell itself, determines the fluidity of the cytosol, and plays an important part in cell division.

- **Microtubules**
- **Intermediate filaments**
- **Microfilaments**
(actin filaments)



Microtubules

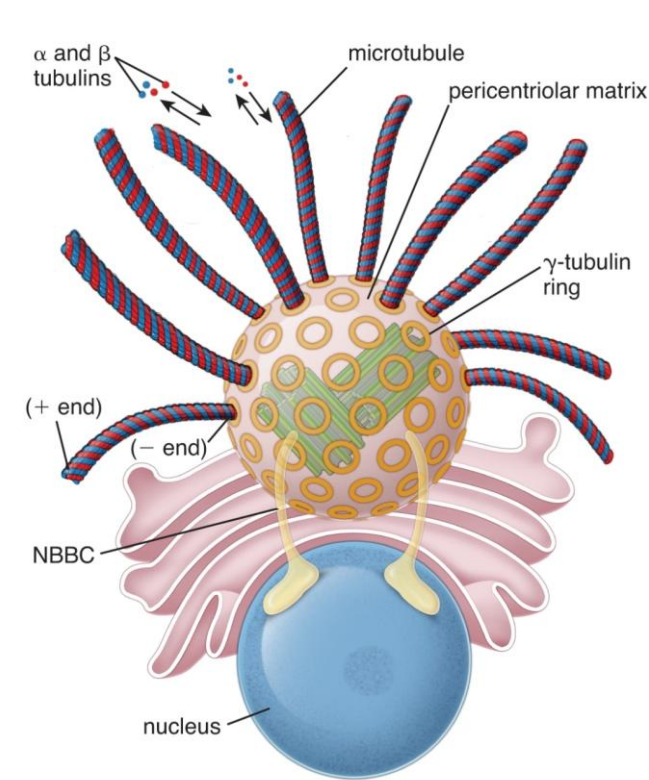
- Hollow cylindrical **tubes** - 25 nm diameter
- Have positively and negatively charged ends = **polar**
- Consist of round (**globular**) protein units (**α - and β -tubulin**)
- 13 linear protofilaments** form 1 microtubule
- Grow from the microtubule organizing center (**MTOC**) by polymerization
 - Requires **ENERGY (GTP)**
- Functions
 - Intracellular transport**
 - Forms the mitotic spindle for cell division**
 - Control chromosomal movement**
 - Form flagella and cilia**



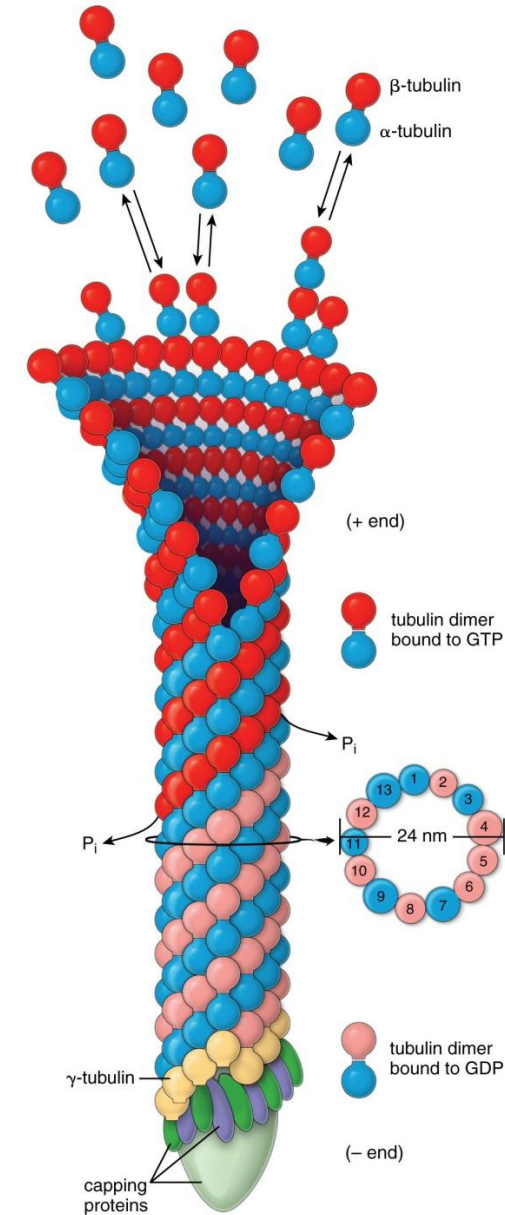
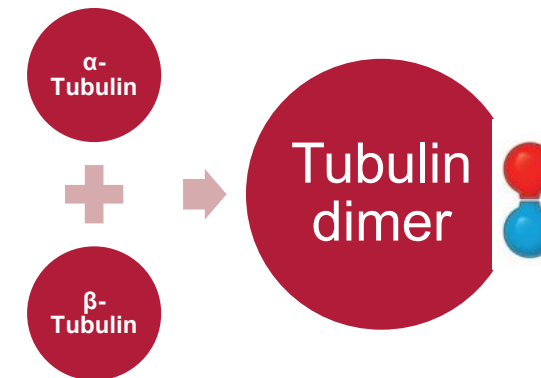
Tau proteins bind to tubulins promoting microtubule assembly and stability

Microtubule Organizing Center (MTOC)

- Centrosome surrounded by protein matrix (**pericentriolar material**)
- Close to the nucleus
- Nucleation sites** for microtubules
 - Gamma (γ) tubulin ring** complexes on the pericentriolar material
 - Initiate microtubule formation (polymerization)
- Microtubule detaches once fully assembled

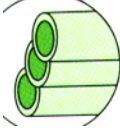


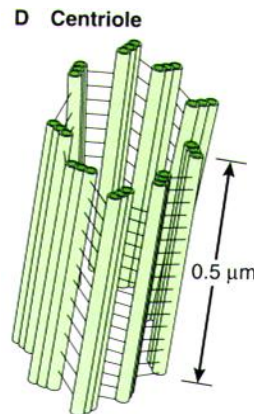
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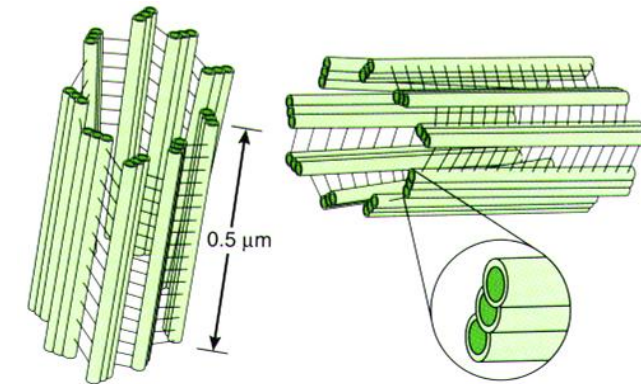
Centriole

- 9 sets of three (**triplets**) microtubules arranged in a circular fashion 
- Connected to each other by **microtubule adherence proteins (MAP)**
- Provide **basal bodies** necessary for assembly of **cilia** and **flagella**
 - centriole attached to cell membrane

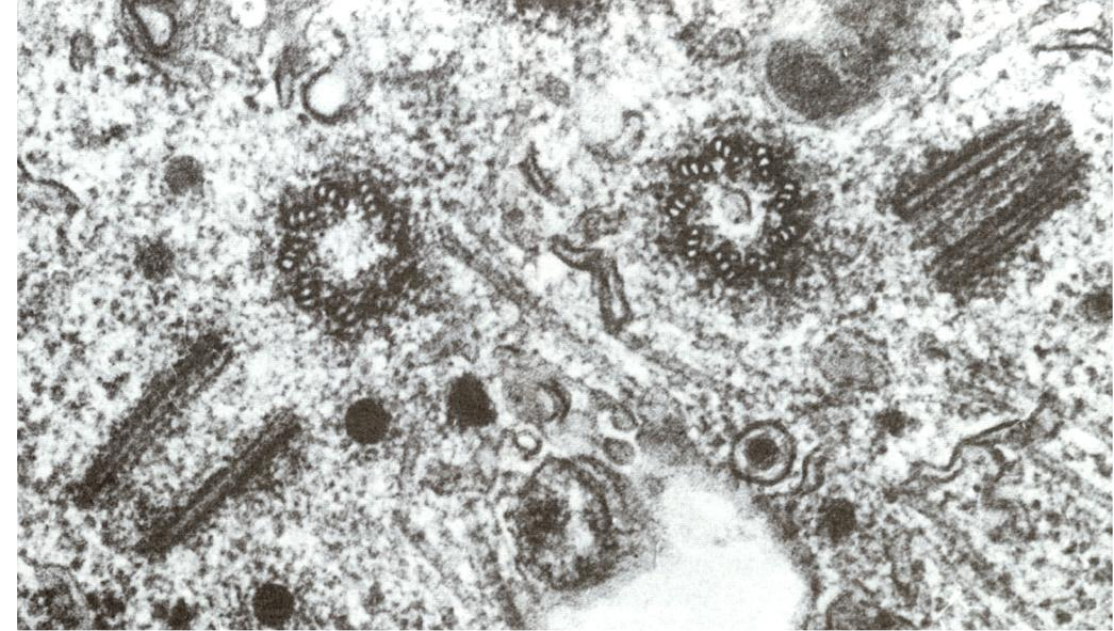
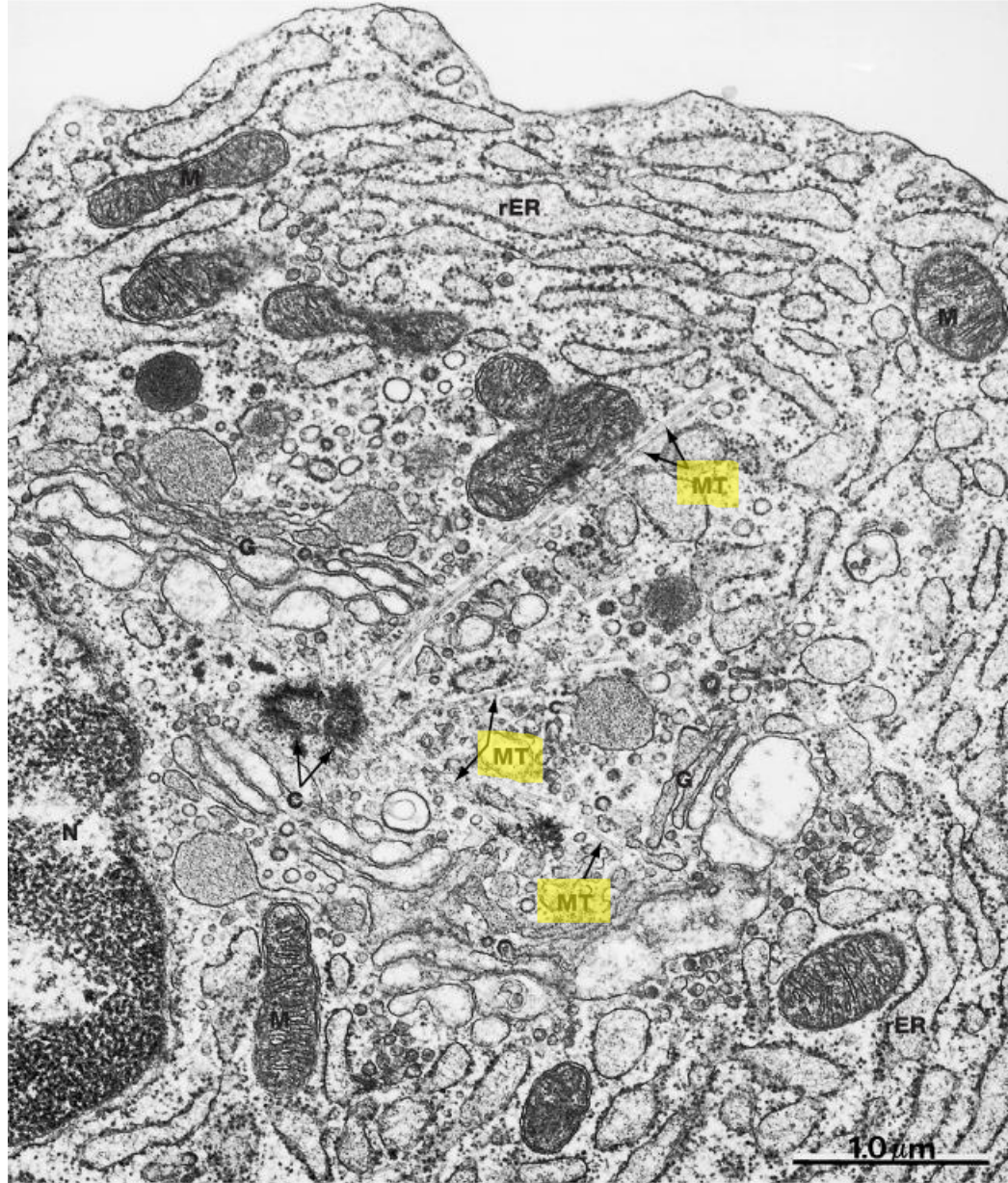


Centrosome

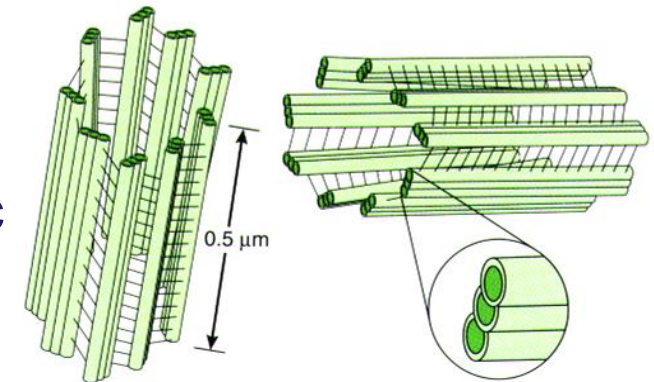
- Two centrioles at 90° angle forms a **centrosome**
- Forms the core of the **MTOC**
- Creates the spindle during mitosis



Microtubules - Electron Micrograph (x30,000 magnification)

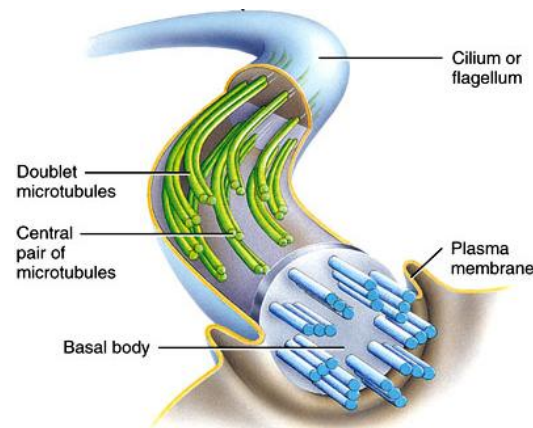
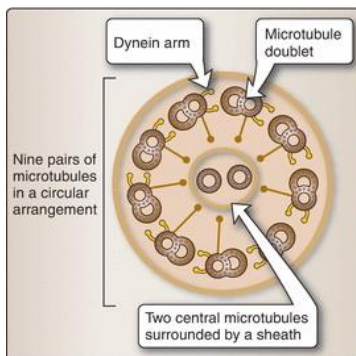


- Centrioles (C)
- Nucleus (N)
- Microtubules (MT)
- Rough endoplasmic reticulum (rER)
- Golgi (G)
- Mitochondria M



Microtubules - Cilia & Flagella

- Highly **specialized motile** apical specializations of cells
- Consist of a core (**axoneme**)
 - 9 sets of two (**doublets**) arranged circularly around 2 centrally placed microtubules (**9+2 arrangement**)
 - Doublets are attached to each other by a pair of **dynein arms** (motor proteins)
- Anchored to the cell via **basal body**
- Movement produced by the bending of the **axoneme**



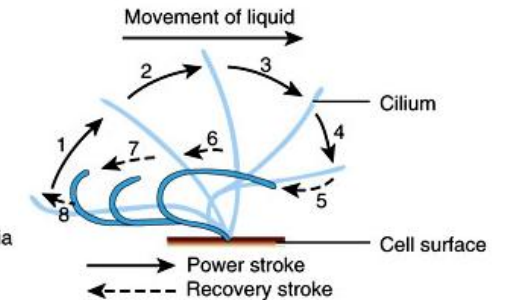
Cilia

Hairlike structures responsible for **active movement** of particles along the surface – found on most epithelium

Movement is rhythmic – active stroke in the direction of movement and slightly less powerful recovery stroke back to starting position



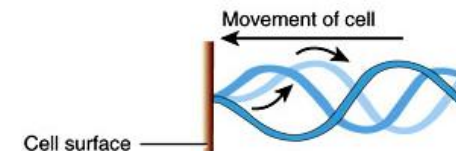
(b) Cilia lining the trachea



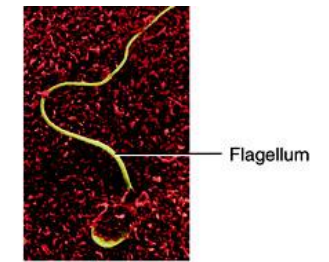
(b) Ciliary movement

Flagella

Long motile tails that **move the entire cell** (e.g. sperm cell)



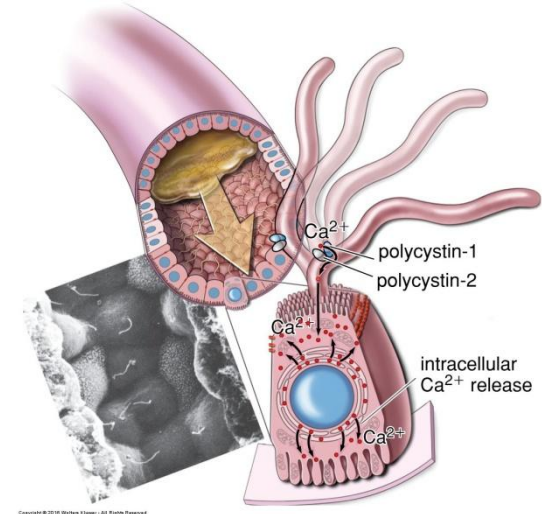
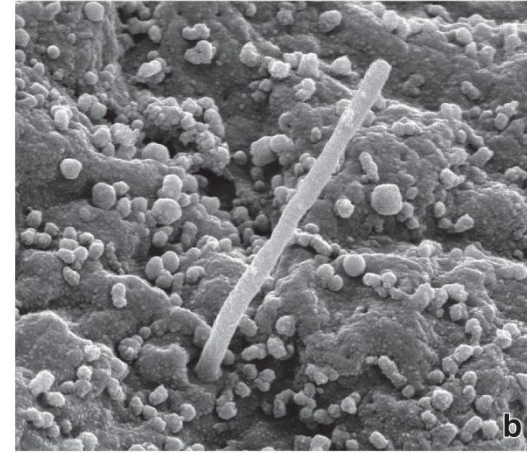
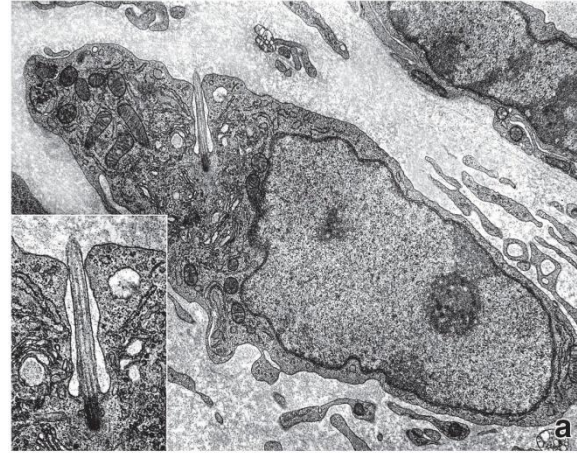
(c) Flagellar movement



(c) Flagellum of a sperm cell

Microtubules - Primary Cilia

- Long **non-motile**, microtubule-based, antennae-like structure
 - 9 sets of two (**doublets**) arranged circularly (**9+0 arrangement**)
- Emanates from almost all cells
- Anchored to cell via the **basal body**
- Functions as **sensory antennae**
 - **Photoreceptors**
 - e.g. outer segment of rods in the eye
 - **Chemoreceptors**
 - e.g. odor detection of olfactory neurons
 - **Mechanoreceptors**
 - e.g. epithelial cells monitors the flow of fluid through the kidney tubules

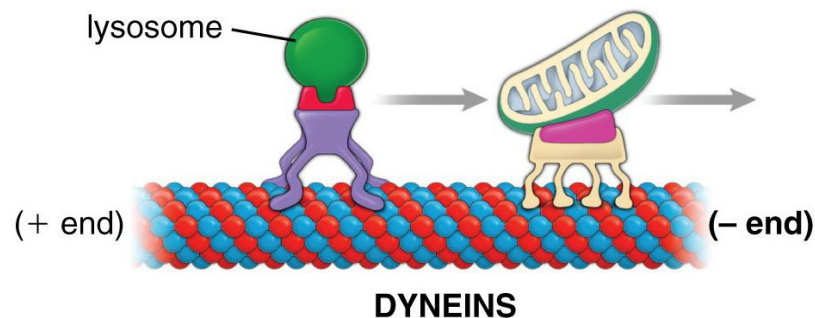


Microtubules – Intracellular Movement

- Vesicles, organelles and microtubules need to move within the cytoplasm to perform their function
- This is done with the aid of microtubules and motor proteins **Dynein and Kinesin**
- **ATP dependent**

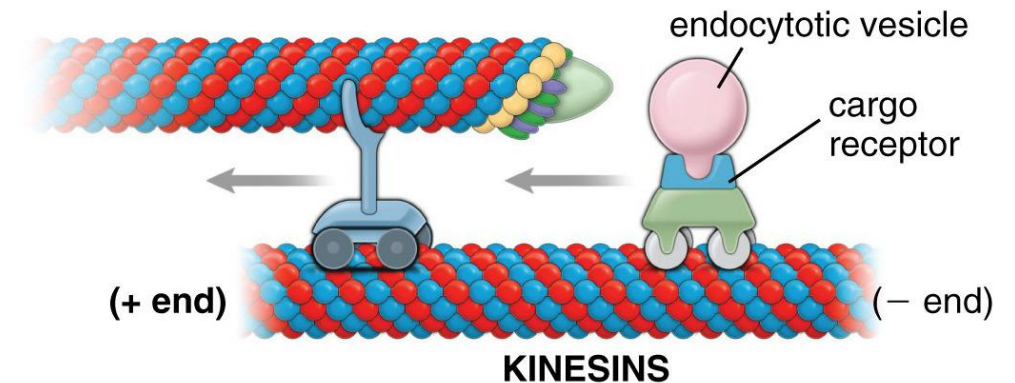
Dynein Family (14 $\mu\text{m}/\text{sec}$)

- Move in the (negative) direction along microtubules (**Retrograde**)
- Largest & fastest of the known molecular motors



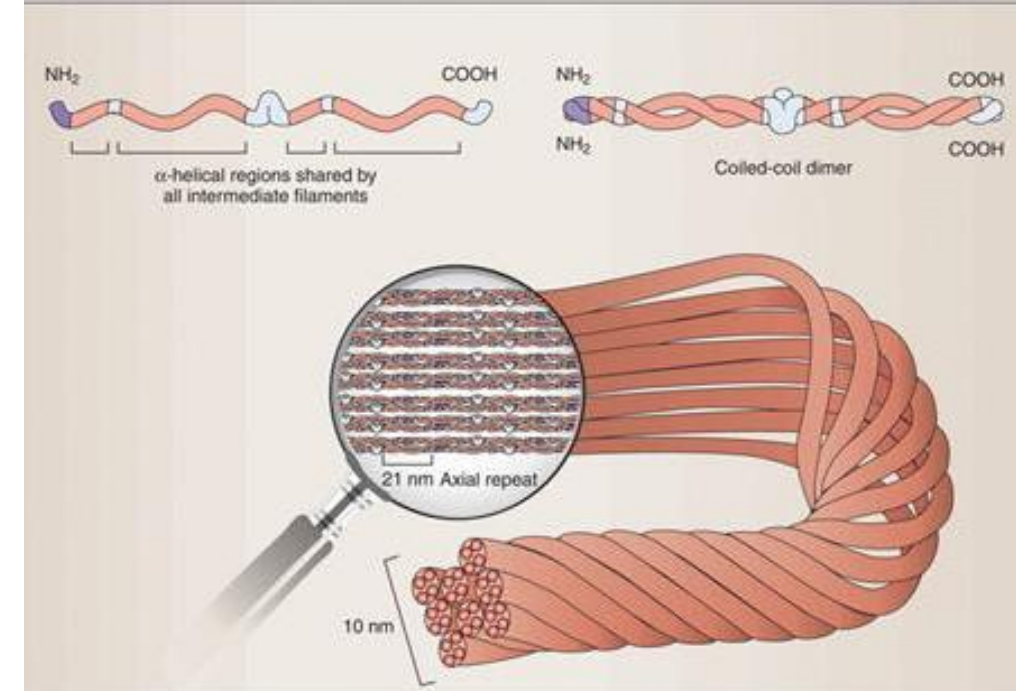
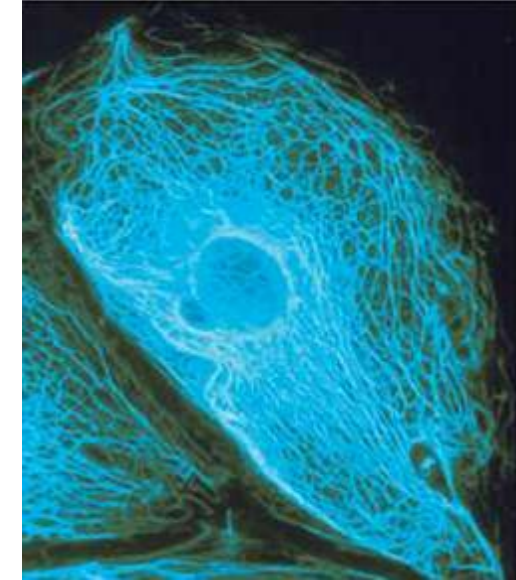
Kinesin Family (3 $\mu\text{m}/\text{sec}$)

- Move in the (positive) direction along microtubules (**Anterograde**)
- ~ 40 distinct kinesins in humans



Intermediate Filaments

- No charge at the ends = **Non-polar**
- α -helical **rod-like protein subunits**
 - Coil together forming rope-like units (8-12 nm diameter)
 - Arranged in a staggered way to increase stability and strength
- Assembled with the help of linking proteins = **No energy required**
- Provide **structural** support
- Essential for integrity of cell-cell & cell-ECM junctions
- 6 types based on their location, structurally similar but their amino- and carboxy-terminal regions are unique to each type
 - e.g. keratin in skin cells

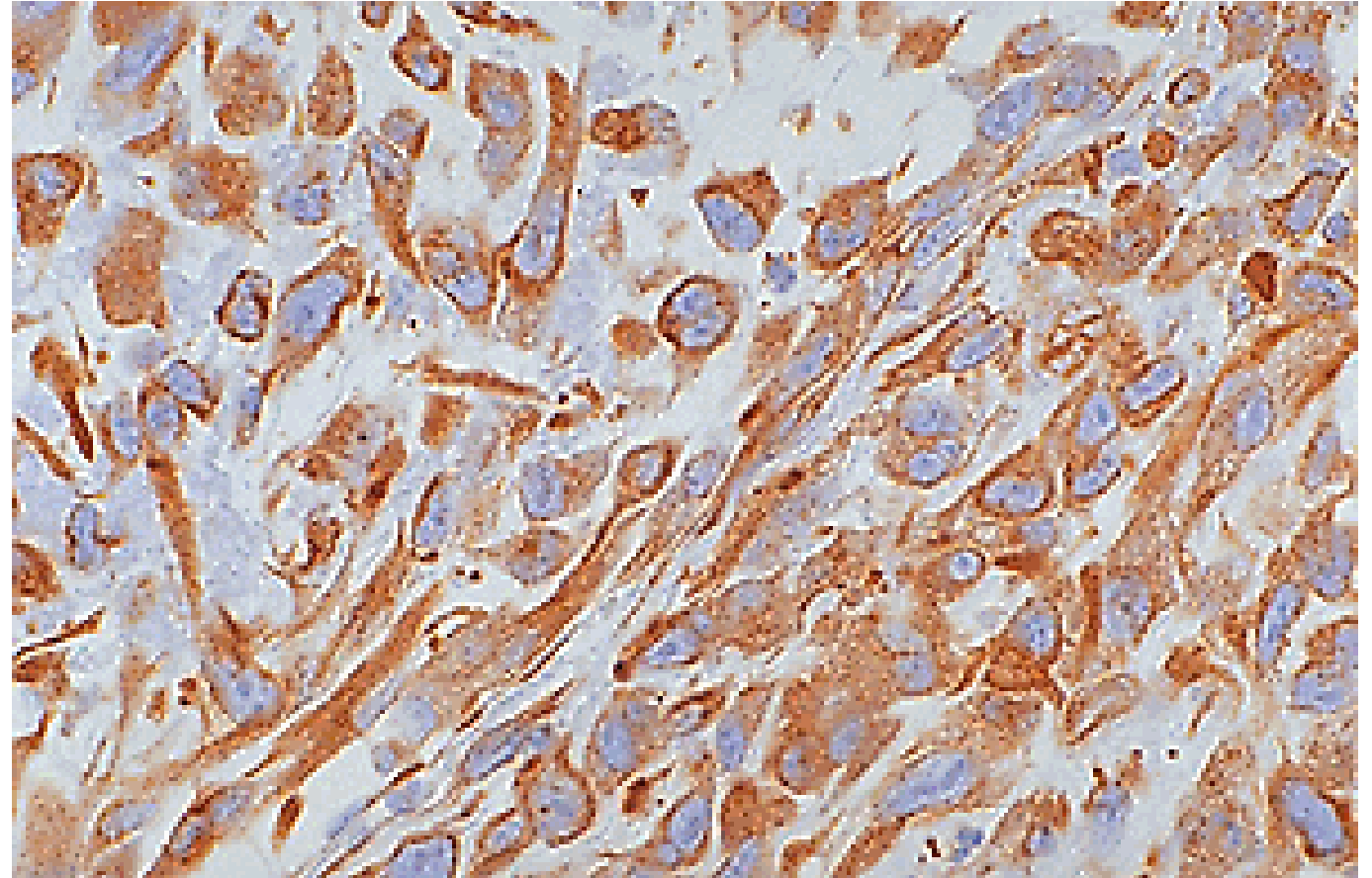


Intermediate Filaments – types and examples

Type	Names	Functions	Examples of where they are found
I & II	Acidic (I) and basic (II) keratins	Form complex network from nucleus to plasma membrane in epithelial cells	Epithelial cells The major component in skin cells and hair
III	Desmin	Support and structure	Muscle cell integrity
	Vimentin		Mesenchymal cells, fibroblast, <u>endothelial cells</u> , <u>macrophages</u> , <u>melanocytes</u> , <u>Schwann cells</u> , and lymphocytes
IV	Neurofilaments, synemin, syncoilin	Protect from mechanical stress and maintain structural integrity in various cell types	Synemin – forms part of focal adhesions Syncoilin – skeletal and cardiac muscle, muscle integrity and force transduction
V	Nuclear lamina (lamins)	Structural role in the nucleus of all cells	Nucleus of every cell in the body
VI	Nestin	Expressed mainly in the nerve cells and is implicated in their growth	Angiogenesis and neuronal progenitor cell marker

Intermediate Filaments – Clinical Correlates

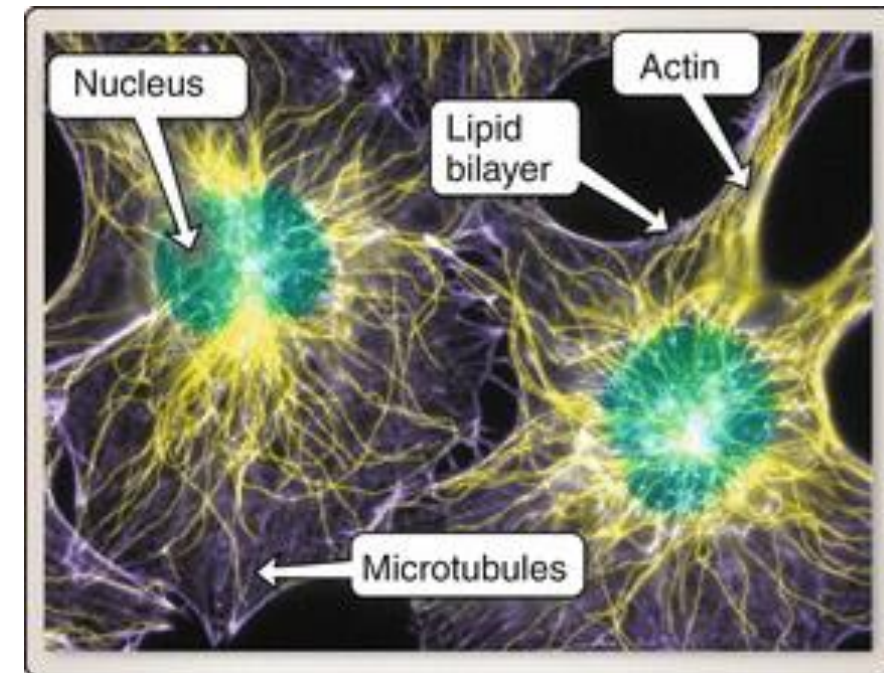
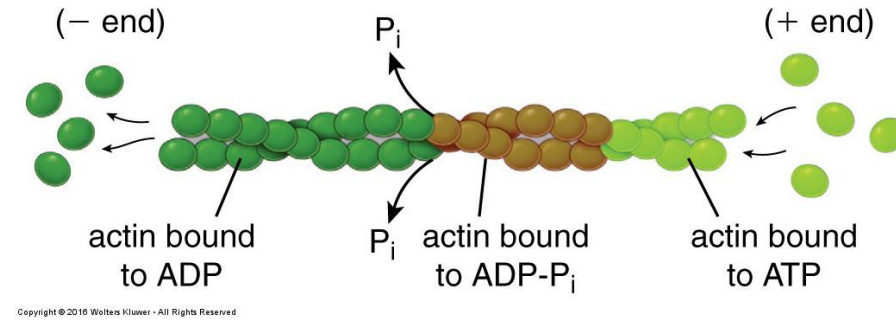
- Special stains can be used to investigate different conditions
- Certain connective tissue dysfunctions can be identified by staining for intermediate filaments
- Useful to identify the origin of secondary tumors



Klatt Edward C. MD, Front Matter, Robbins and Cotran Atlas of Pathology (Fourth Edition), edited by Klatt Edward C. MD, 2021, Pages iii-iii, ISBN 978-0-323-64018-3, <http://dx.doi.org/10.1016/B978-0-323-64018-3.01001-2>.
(<https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323640183010012>)

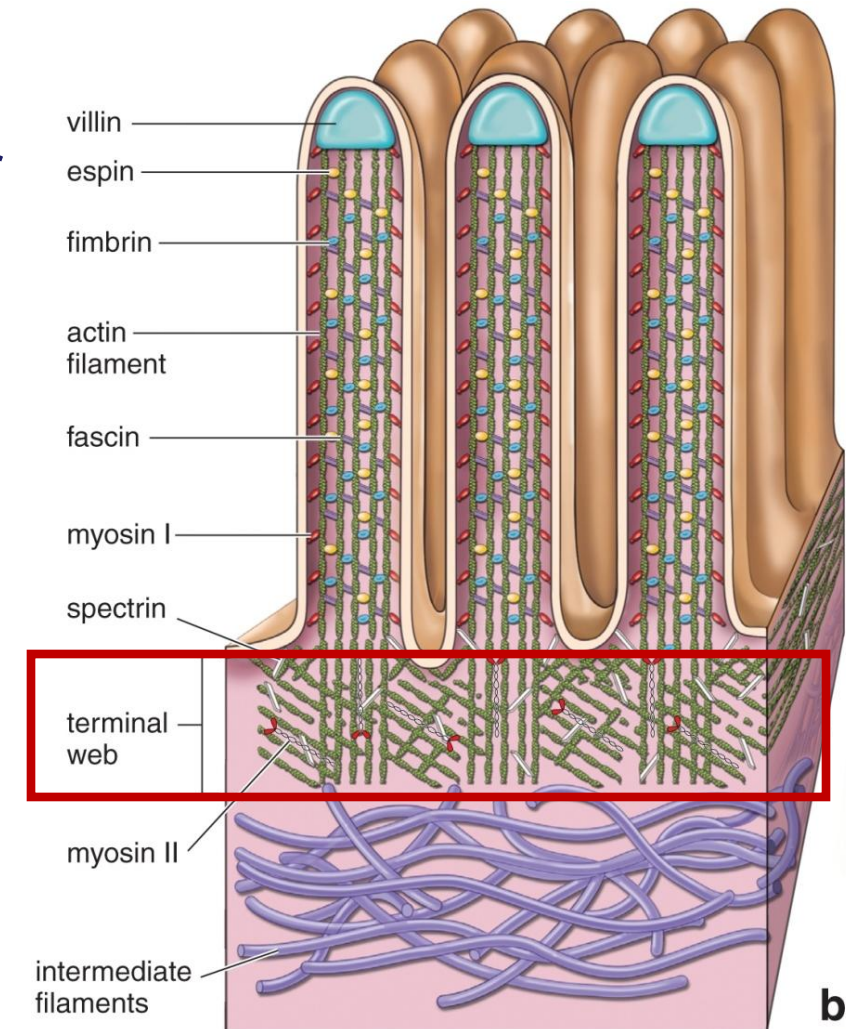
Microfilaments (Actin Filaments)

- Consist of round proteins = **actin molecules**
 - Free actin molecules in the cytoplasm: **G-actin**
 - Polymerized actin in a filament: **F-actin**
- May exist as single **filaments (7 - 8 nm diameter)**, in bundles, or in networks
- Found in most cells
- Helps to establish a **cytoplasmic protein framework**
- Have positive and negative charged ends = **polar**
- Nucleation site: **3 G-actin** molecules each with a bound **ATP**
- Requires ATP for polymerization



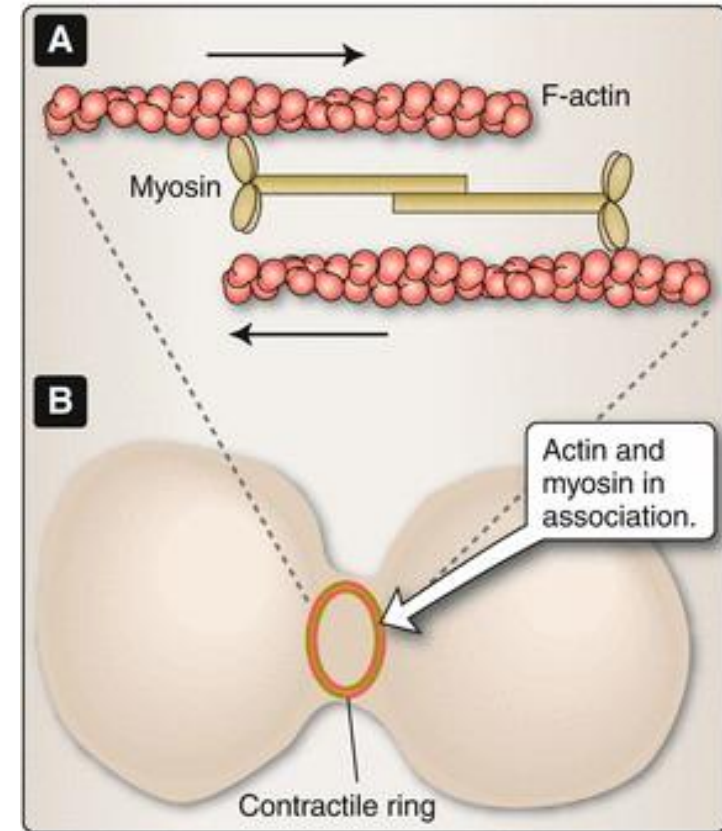
Microfilaments (Actin Filaments) – Functions

- Anchorage and movement of membrane proteins
 - **Terminal Web**
 - Provides frameworks for the cell, attachment for microvilli and basal bodies
 - Also found on lateral surface where cell to cell adhesions are
- Structural core of **microvilli & stereocilia**
- Extension of cell processes
 - e.g. **phagocytosis and cell movement**
- **Cell motility:** locomotion without flagella
- Controls the **viscosity of the cytosol**
 - Aggregation of actin increases the viscosity while dispersion decreases it



Actin motor proteins - Myosin

- Also referred to as **thick filaments**
- Have a round head domain that interacts with F-actin and a tail containing an ATP-binding site
- Myosin tail can bind to cellular structures and pull them along the actin filament
- It helps to stiffen and prevent deformation of the plasma membrane
- Responsible for **closure of the gaps in wounds**
- Responsible for formation of the “pinch” during cell division
- Plays an active role in skeletal muscle contraction

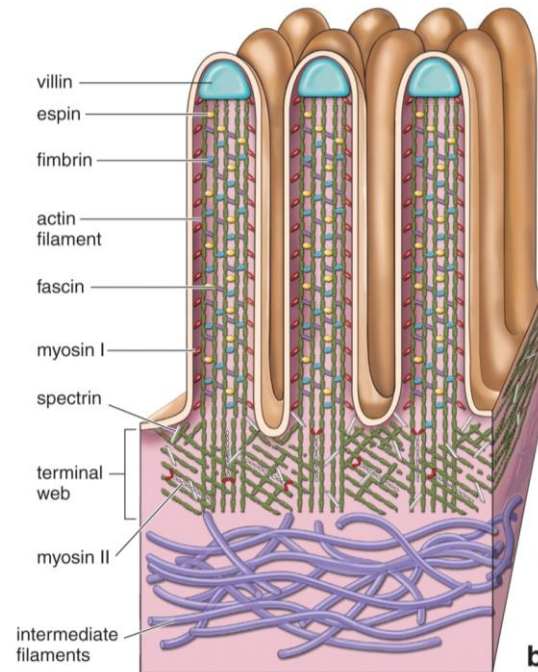


Microfilaments – Microvilli and Stereocilia

- Cytoplasmic extensions on the apical surface of some epithelium that increase the surface area for absorption
- Non-motile
- Consist of a **core of 25-30 actin filaments** anchored to the terminal web

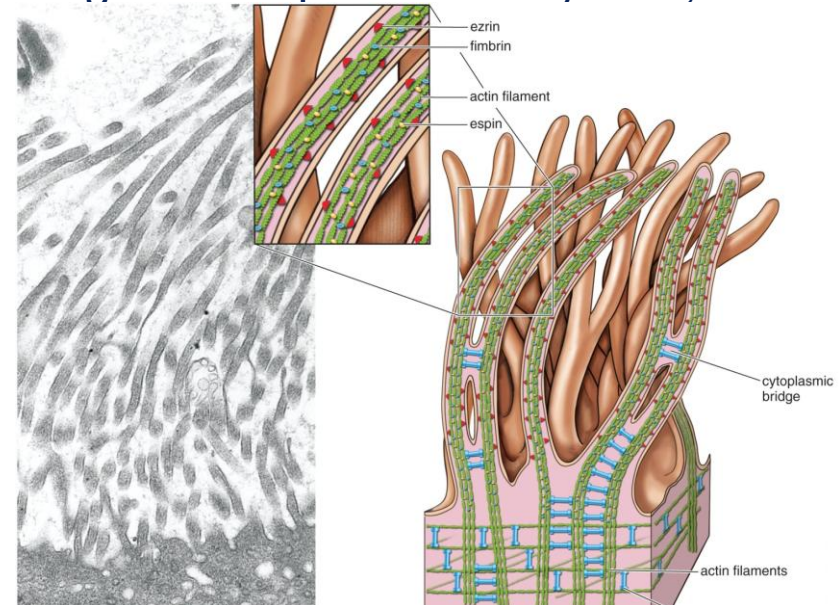
Microvilli

- Cylindrical, membrane-bound cytoplasmic projections: 1-3 μm long



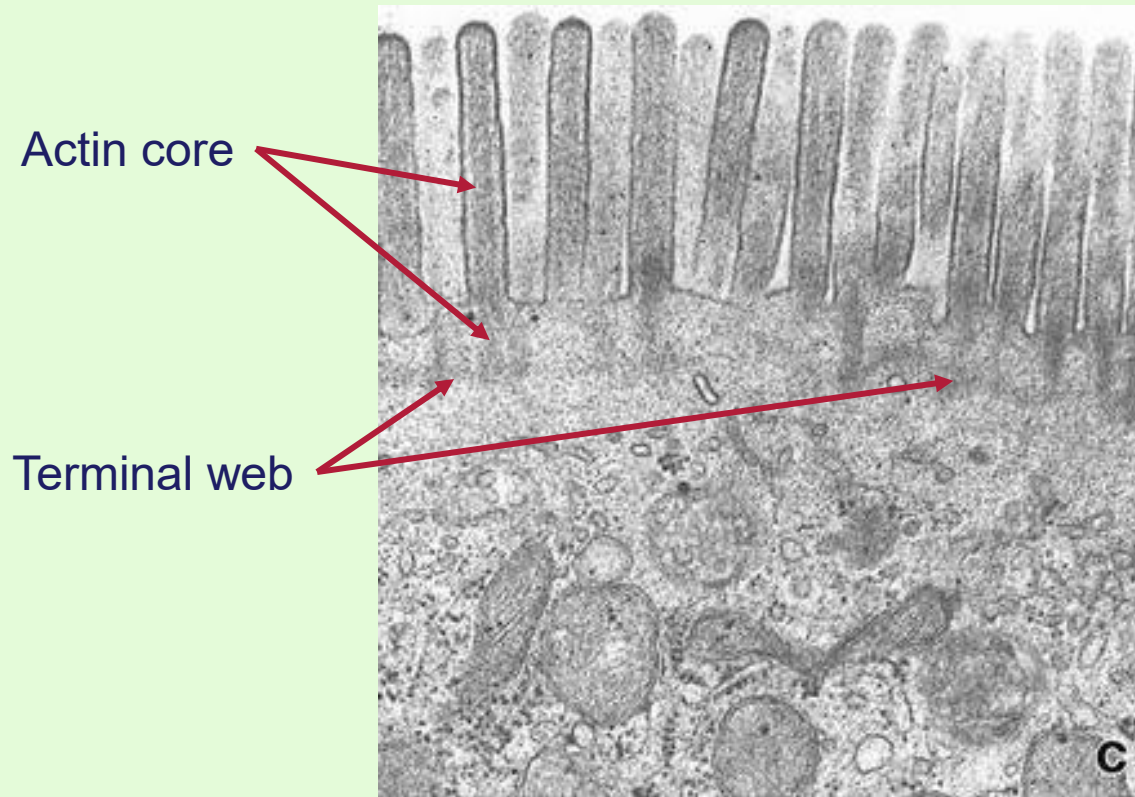
Stereocilia

- Unusually long microvilli up to 120 μm long
- Limited distribution
 - e.g. male reproductive system, ear

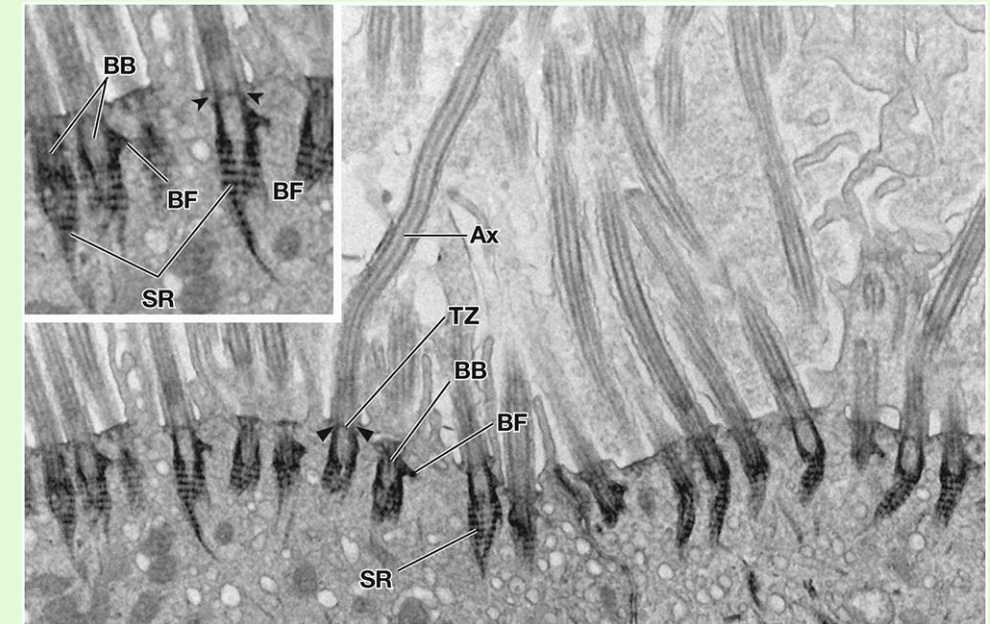


Comparison between microvilli and cilia

microvilli

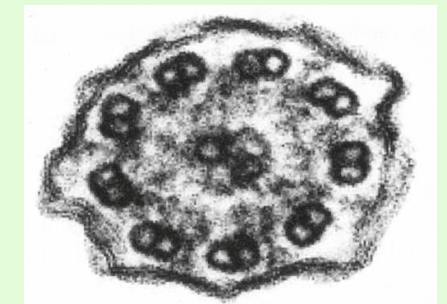


cilia



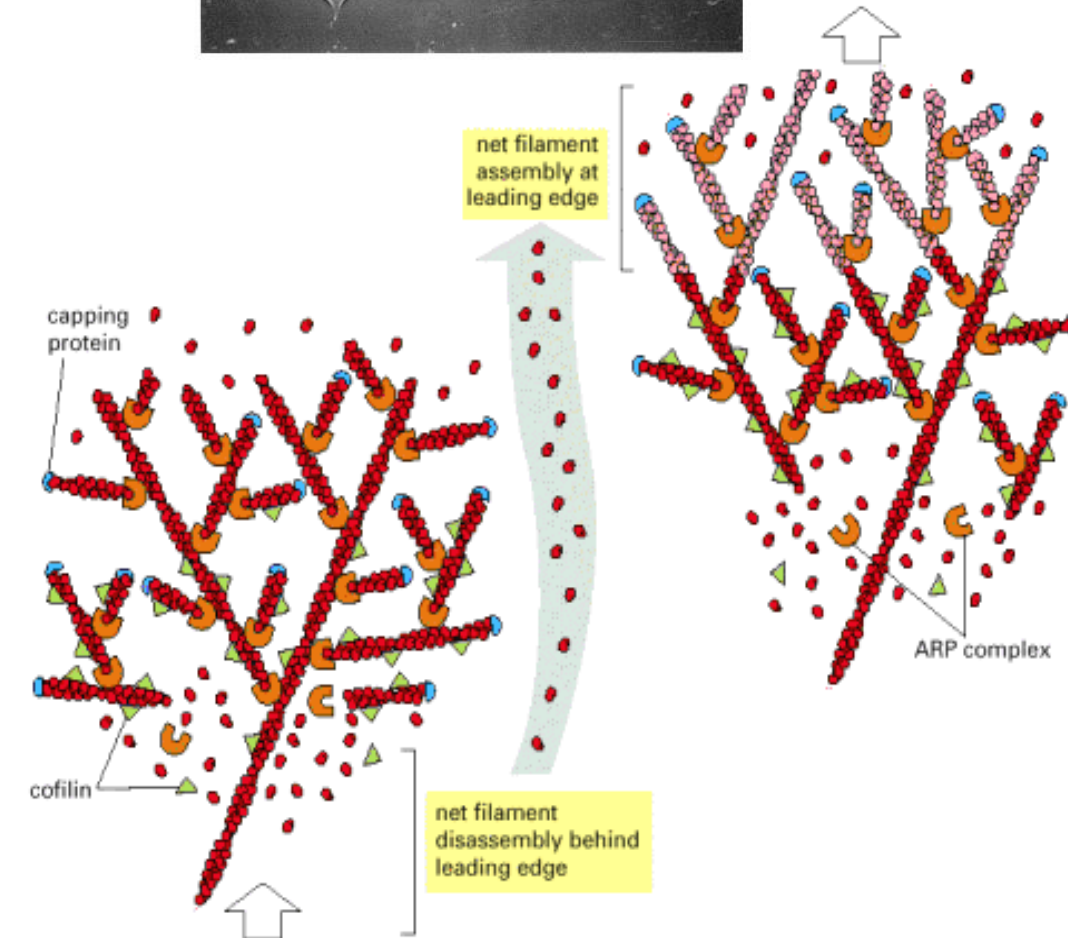
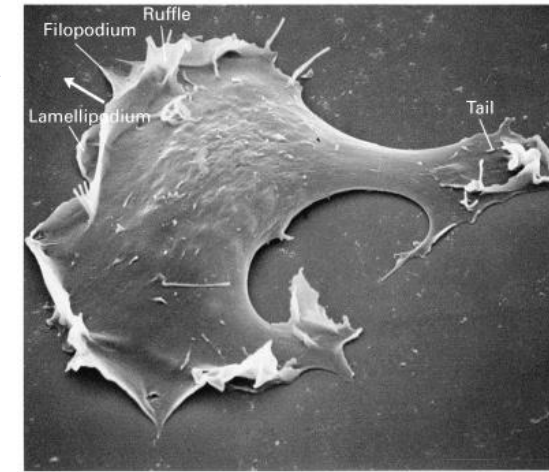
Basal bodies (*BB*)

Axoneme (*Ax*), (9 + 2 microtubular arrangement)



Microfilaments – Cell movement

- Signaling molecules bind to surface receptors
 - activates second messengers
 - induces actin polymerization toward the signal: **chemotaxis**
- Actin web undergoes treadmilling
 - Assembling toward the signal
- The polymerization produces plasma membrane protrusions
 - **Filopodia**
 - Finger-like projections
 - **Lamellipodia** (e.g. fibroblasts)
 - Sheet-like structures
 - **Pseudopodia** (e.g. white blood cells)
 - 3 dimensional projections
- Myosin accumulates at the other end of the cell
 - Myosin-actin interaction allows for complete movement of the cell



Cytoskeleton – Clinical Correlates

- Some chemicals and poisons can impair the function of the cytoskeletal components
- Changes can alleviate inflammatory conditions, downregulate excessive cell division, or be fatal

Microtubule poisons

- Colchicine
 - Treatment of gout
 - Binds to unpolymerized tubulin molecules
 - Prevents polymerization
 - Prevents release of inflammatory initiators
- Vinblastine and Vincristine
 - Inhibit formation of mitotic spindle
 - Inhibit uncontrolled cell division
- Taxol® (Paclitaxel)
 - Stabilizes and prevents microtubule disassembly
 - Arrests dividing cells in mitosis



Actin Toxins

- Phalloidin (toxin found in *Amanita phalloides*)
 - Binds F-actin more tightly than G-actin
 - Promotes excessive polymerization and inhibits depolymerization
 - Inhibits cell movement
- Cytochalasins
 - Block polymerization of actin
 - Can be used to inhibit cell movement and division & induce programmed cell death

Amanita phalloides -
“death caps” or “angels of death”

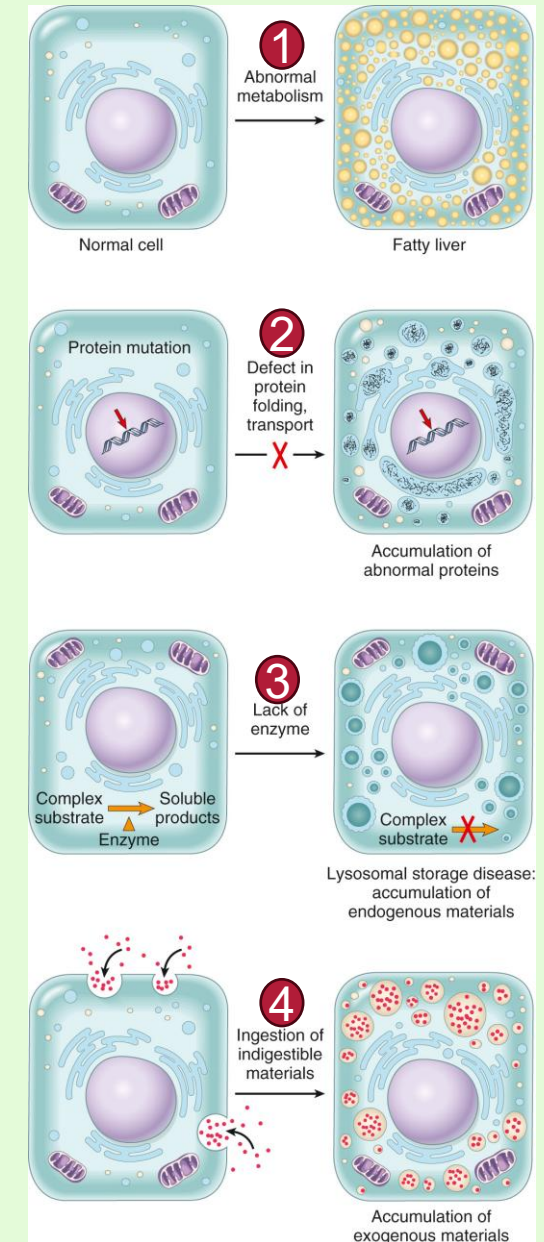


Cell Inclusions - Mechanisms

- **Cytoplasmic or nuclear structures formed from metabolic products of the cell**

- Normal cell inclusions occur in most cells
- Abnormal cell inclusions is a manifestation of metabolic derangements

- 1 *Inadequate removal* of a normal substance
 - secondary to defects in packaging and transport
 - fatty change (steatosis) in the liver
- 2 Accumulation of an endogenous substance
 - genetic or acquired *defects in its folding, packaging, transport, or secretion*, as with certain mutated forms of α_1 -antitrypsin
- 3 *Failure to degrade* a metabolite
 - inherited enzyme deficiencies, typically lysosomal enzymes.
 - *lysosomal storage diseases*
- 4 Deposition and accumulation of an *abnormal exogenous substance*
 - cell has neither the enzymatic machinery to degrade the substance nor the ability to transport it to other sites
 - carbon or silica particle accumulation



Normal Cell Inclusions

Functional

Glycogen (storage form of glucose)

- Non-membrane-bound, TEM dense bodies
- Catabolism releases glucose for energy

Lipids (energy store and source of short carbon chains for membrane synthesis)

- Non-membrane bound, TEM dense, Fat droplets
- Spherical droplets of triglyceride
- Liquid at body temperature

Byproducts from Metabolism

Lipofuscin (brownish-gold pigment)

Accumulation of oxidized lipids, phospholipids, metals and other organic molecules as result of lysosomal digestion
Accumulates over years (“wear & tear” pigment)

Melanin

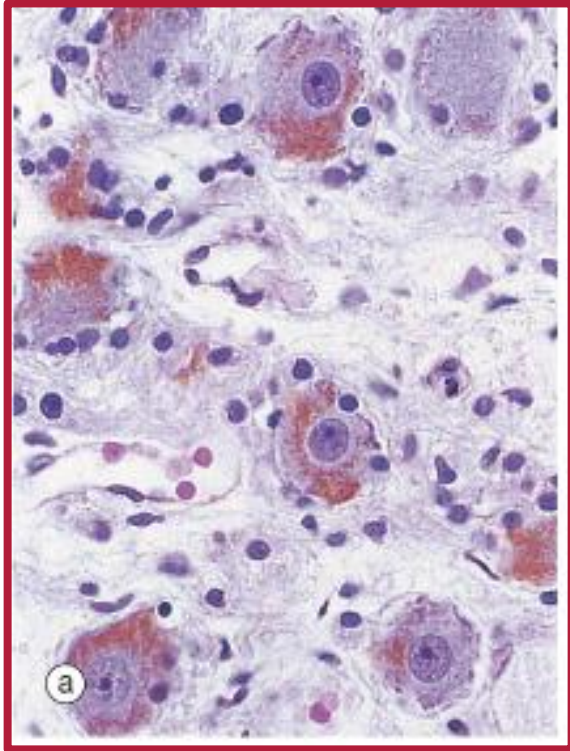
Dark brown/brown/reddish pigment produced by the oxidation of tyrosine
Melanocytes in the skin & responsible for color of skin/hair pigmentation
Also produced by certain neurons of the brain

Hemosiderin

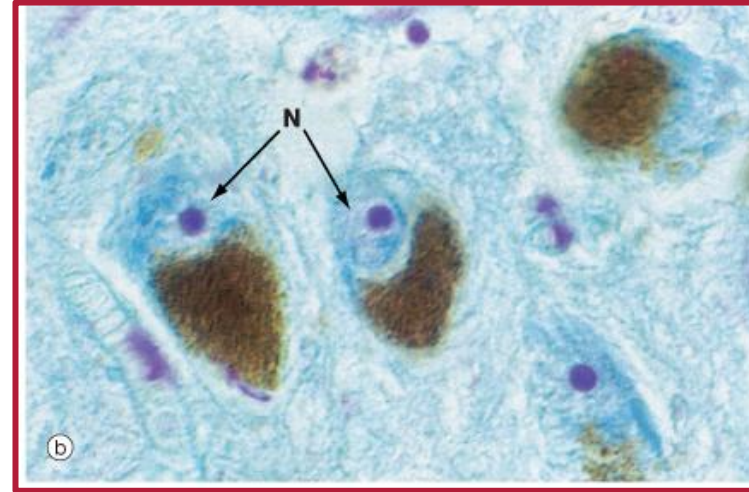
Iron-storage complex in cytoplasm
Due to phagocytosis of red blood cells
Mostly found in spleen

Inclusions

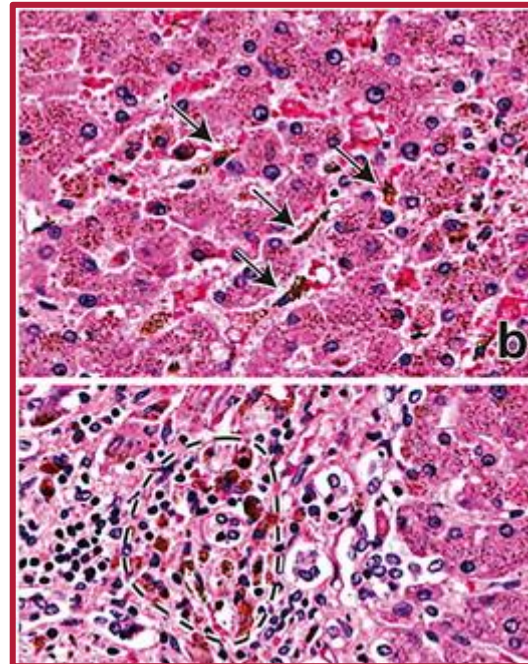
Lipofuscin



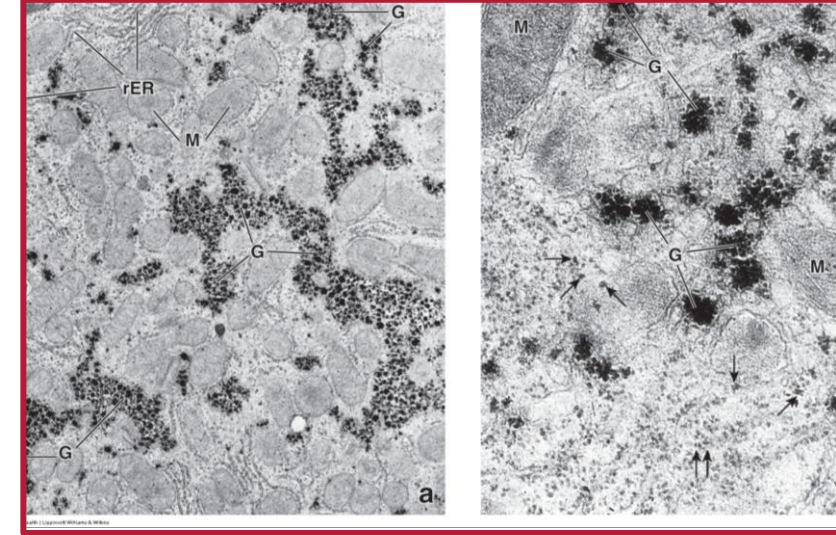
Melanin



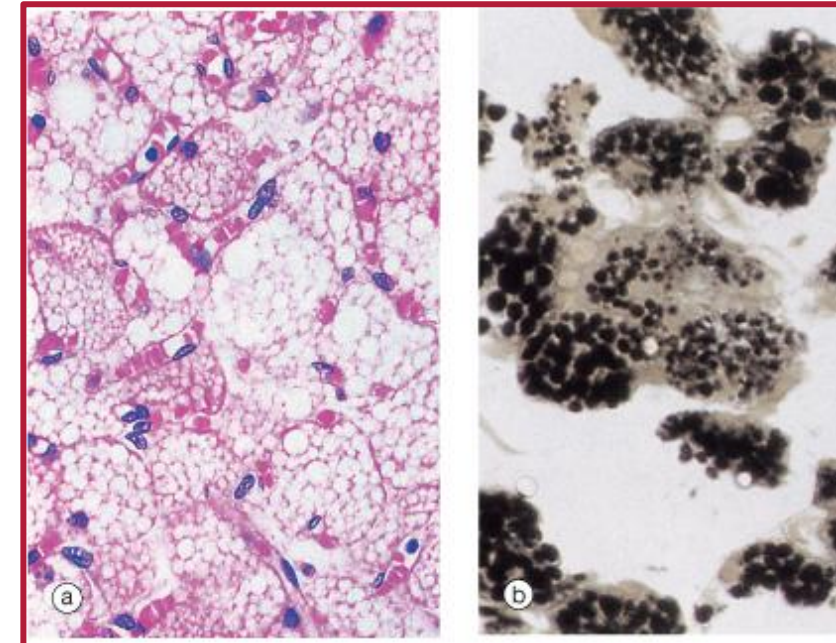
Hemosiderin



Glycogen



Lipid



fin